ALM-3

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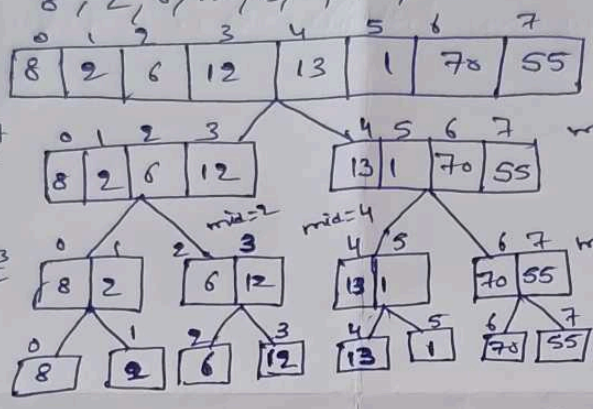
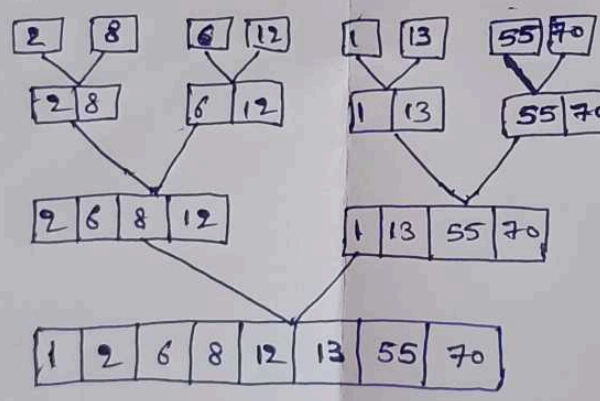
Design And Analysis Of Algorithms

Merge sort - (divide & conquer)

1) sort the given elements using merge sort

8, 2, 6, 12, 13, 1, 70, 55





Merge sort divides the given array until single elements using recursive calls and again joins the array by doing sorting. It used mid to divide

array where

~~mid = (first + last)/2~~

$$mid = first + \frac{(last - first)}{2}$$

Hence it is called Divide & Conquer method.

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ALM-4

DAA

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1) n=7 (Job Sequence with deadlines)

Jobs	J1	J2	J3	J4	J5	J6	J7
profits	35	30	25	20	15	12	5
deadlines	3	4	4	2	3	1	2

Time	Job	Deadlines	profit
0	0	0	0
1	[J4]	1	20
2	[J4, J3]	2	45
3	[J4, J3, J1]	3	80
4	[J4, J3, J2, J1]	4	110

Job sequence ⇒ [J4, J3, J1, J2]

[J3, J4, J1, J2]

0 ⟷ 1 ⟷ 2 ⟷ 3 ⟷ 4

J4 J3 J1 J2

20 + 25 + 35 + 30

⇒ 110

In job sequence with deadlines, we need to select the jobs within the given deadlines, so that the profit is maximum.